

Motivation

- A deputy spacecraft is tasked with an on-orbit inspection of a chief satellite while navigating the orbital environment under constraints.
- GPS-denied navigation - The deputy spacecraft must rely on onboard sensors for self-localization.
- Field-of-view (FOV) constraints - Limits on the observable area that a sensor or camera can capture.
- Illumination constraints - The deputy spacecraft can only inspect illuminated features within its FOV, as the Sun is the sole light source.
- Avoidance regions - The deputy spacecraft must not collide with or drift too far from the chief satellite.

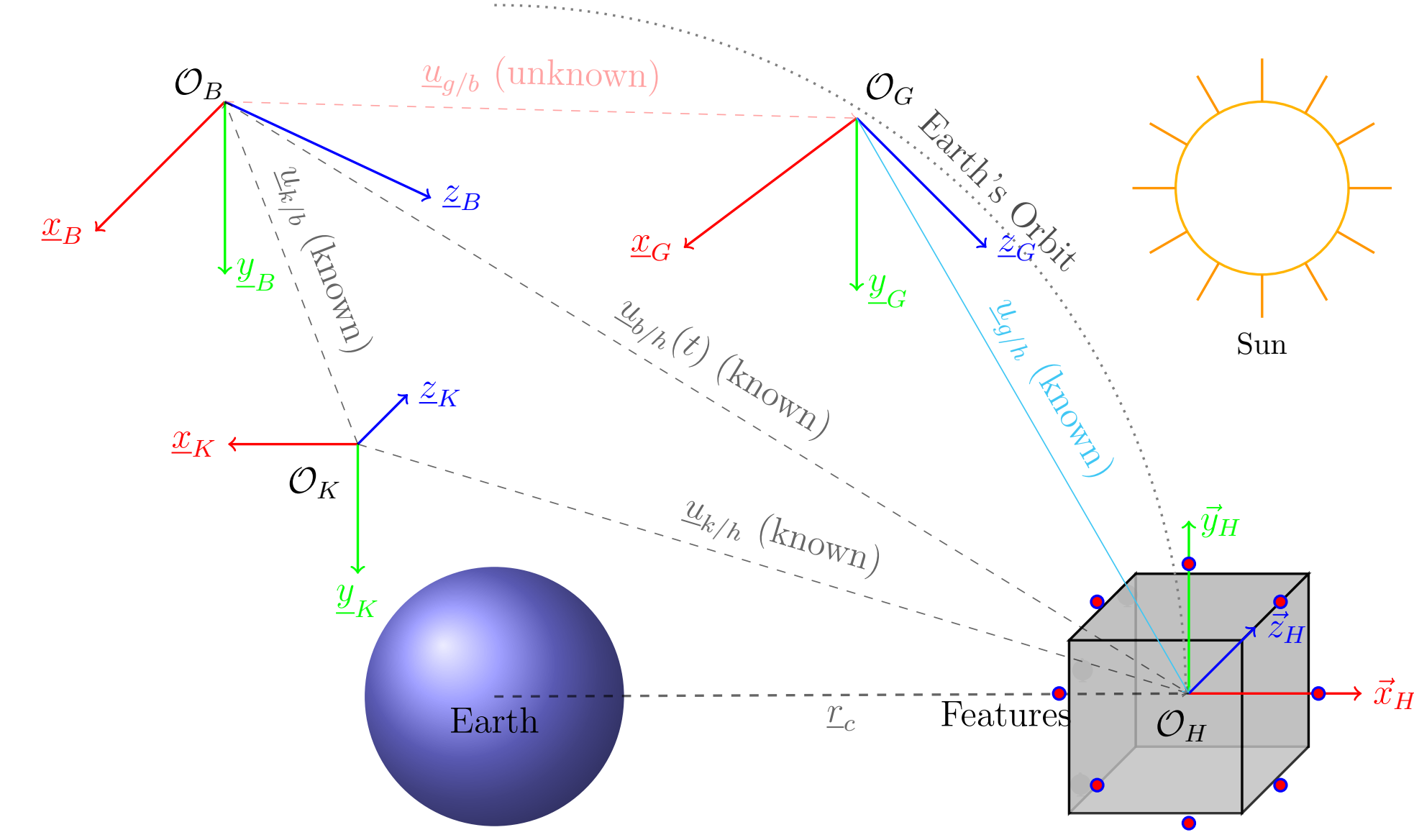


Fig. 1: On-orbit satellite inspection in Hill's frame

Literature Review

- Earth-orbiting spacecraft typically use GPS and ground-based tracking, while deep-space missions rely on the Deep Space Network (DSN) and celestial navigation methods, such as optical planet detection and X-ray pulsar navigation [1].
- Recent approaches to autonomous satellite inspection include information-driven guidance using Bayesian-based metrics [2], photogrammetry with extended Kalman filters [3], and reinforcement learning-based control [4].
- Image-based observers [5] can reconstruct 3D structures for visual servo control, but challenges such as feature occlusions and limited field-of-view affect tracking performance.
- While image-based control techniques, such as those presented in [6], can be used to adjust camera focal length to maintain target visibility, their approach is not feasible for low-cost cameras with fixed focal lengths.

Goal

- Use pixel coordinates of features on the chief satellite to create a local map of the environment for relative navigation.
- Maximize *information gain* from a camera fixed to the deputy spacecraft.
- Ensure the camera motion maximizes the number of illuminated points within the FOV of the camera.
- Prevent the deputy spacecraft from colliding or straying too far from the chief satellite.

Problem Formulation

- Let $\mathbf{x}(t) := \begin{bmatrix} \underline{p}_{g/b}(t)^\top & \underline{v}_{g/b}(t)^\top \end{bmatrix}^\top \in \mathbb{R}^6$, where $\underline{p}_{g/b}(t)$ and $\underline{v}_{g/b}(t)$ represent the relative position and velocity of the goal location relative to the deputy spacecraft, respectively, coordinatized in $\mathcal{F}_H$. The linearized Clohessy-Wiltshire (CW) dynamics is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (1)$$

$$\text{where } \mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \frac{1}{m} \begin{bmatrix} 0_{3 \times 3} \\ \mathbf{I}_{3 \times 3} \end{bmatrix}, \quad \mathbf{u}(t) = \begin{bmatrix} (F(t))_x & (F(t))_y & (F(t))_z \end{bmatrix}^\top \in \mathcal{U} \subset \mathbb{R}^3,$$

n is the mean motion of the chief satellite, $F(t) \in \mathbb{R}^3$ is the net force applied by the thrusters on the center of mass of the deputy spacecraft, and m is the mass of the deputy spacecraft.

- The finite horizon continuous-time optimal control problem is

$$\begin{aligned} \min_{\mathbf{u} \in \mathcal{U}} \quad & \int_0^{t_f} r(\mathbf{x}(\tau), \mathbf{u}(\tau)) \, d\tau + V(\mathbf{x}(t_f), t_f), \\ \text{s.t.} \quad & c(\mathbf{x}(t), \mathbf{u}(t), t) \leq 0, \end{aligned} \quad (2)$$

subject to the dynamic constraints in (1).

- Running cost - $r(\mathbf{x}(t), \mathbf{u}(t)) = \mathbf{x}^\top(t) \mathbf{Q} \mathbf{x}(t) + \mathbf{u}^\top(t) \mathbf{R} \mathbf{u}(t) + \gamma_c \langle C \mathbf{x}(t), \underline{u}_{g/h} \rangle^2$, where $\mathbf{Q} \succ 0$, $\mathbf{R} \succ 0$, $\mathbf{C} = \begin{bmatrix} 0_{3 \times 3} & \mathbf{I}_{3 \times 3} \end{bmatrix}$, and $\gamma_c > 0$ is the orthogonality penalty.
- Barrier constraint - $c(\mathbf{x}(t), \mathbf{u}(t), t) = \dot{\Psi}(\mathbf{x}(t), \mathbf{u}(t), t) - \alpha(h_r(\mathbf{x}(t), t))$, where $\alpha : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a class $\mathcal{K}$ function, $\Psi(\mathbf{x}(t), t) = \frac{k_b}{h_r(\mathbf{x}(t), t)}$ is the robust barrier function, $h_r(\mathbf{x}(t), t)$ is the keep-in-zone and keep-out-zone constraint, and $k_b > 0$ is the barrier penalty gain.

Image-based Distance Observer

- Let $\theta_i(t) = [r_{b/h_i}(t) \, r_{k/b}(t) \, r_{k/h_i}(t)]^\top \in \mathbb{R}^3$ be a concatenated vector of the unknown distances, where the scalars $r_{b/h_i}(t) > 0$, $r_{k/b}(t) > 0$, and $r_{k/h_i} > 0$ are the unknown Euclidean distances between the deputy spacecraft, key frame, chief, and the i -th feature of the chief satellite.
- Let $\phi_i(t) = \begin{bmatrix} \underline{u}_{b/h_i}^\top(t) \\ \underline{u}_{k/b}^\top(t) \end{bmatrix} \underline{v}_{b/h}(t)$ and let $r_{b/h_i}^f(t)$ and $r_{k/b}^f(t)$, and $\phi_i^f(t)$ be the filtered form of the $r_{b/h_i}(t) > 0$, $r_{k/b}(t) > 0$, and $\phi_i(t) \in \mathbb{R}^2$.
- Let $\mathcal{Y}_i(t) = (Y_i^\top(t) Y_i(t))^{-1} Y_i^\top(t) \underline{u}_{k/h_i} \in \mathbb{R}^2$, $\mathcal{U}_i(t) = \lambda \phi_i^f(t) + \begin{bmatrix} r_{b/h_i}^f(t) \\ r_{k/b}^f(t) \end{bmatrix} \in \mathbb{R}^2$, where $Y_i(t) = [\underline{u}_{b/h_i}(t) \, -\underline{u}_{k/b}(t)] \in \mathbb{R}^{3 \times 2}$, $\Lambda(s) = \lambda s + 1$ and $s = \frac{d(\cdot)}{dt}$ is the complex variable of Laplace transform.
- Let $\hat{\theta}_i(t) \in \mathbb{R}^3$ be the concatenated vector of the distance estimates updated using

$$\dot{\hat{\theta}}_i(t) = \begin{cases} 0, & \sigma_i = u, \\ \text{proj} \left(\mu_i(t) + k_\theta \mathcal{Y}_i^\top(t) \left(\mathcal{Z}_i(t) - \mathcal{Z}_i(t) \hat{\theta}_i(t) \right) \right), & \sigma_i = a, \end{cases} \quad (3)$$

where $\mu_i(t) = [\phi_i(t)^\top \, 0]^\top \in \mathbb{R}^3$, $\mathcal{Z}_i(t) = \begin{bmatrix} Y_i(t) \, 0_{3 \times 1} \\ 0_{1 \times 2} \, 1 \end{bmatrix} \in \mathbb{R}^{4 \times 3}$, $\mathcal{Z}_i(t) = \begin{bmatrix} \underline{u}_{k/h_i} \Sigma_{\mathcal{Y}_i}^{-1} \mathcal{U}_i(t) \\ \Sigma_{\mathcal{Y}_i}^{-1} \mathcal{U}_i(t) \end{bmatrix} \in \mathbb{R}^4$, $\Sigma_{\mathcal{Y}_i} = \sum_{j=1}^M \frac{\mathcal{Y}_i^\top(t_j) \mathcal{Y}_i(t_j)}{1 + \|\mathcal{Y}_i(t_j)\|^2}$, and $k_\theta \in \mathbb{R}^{3 \times 3}$ is a constant gain matrix.

Assumption: Sufficient Camera Motion

The camera has sufficiently rich motion so that there exists a set of features $\mathcal{A}(t) \subset \mathcal{N}$, a constant $\underline{\Sigma} > 0$, and a set of finite time instances $\mathcal{T} = \{t_{i,j}^*\}_{i \in \mathcal{A}(t)} > \varsigma$ such that for all $t \geq t_{i,j}^*$ with $t_{i,j}^* \in (t_{i,j}^a, t_{i,j}^u)$, $j \in \mathbb{N}$,

$$\lambda_{\min}(\Sigma_{\mathcal{Y}_i}) > \underline{\Sigma}. \quad (4)$$

Simulation

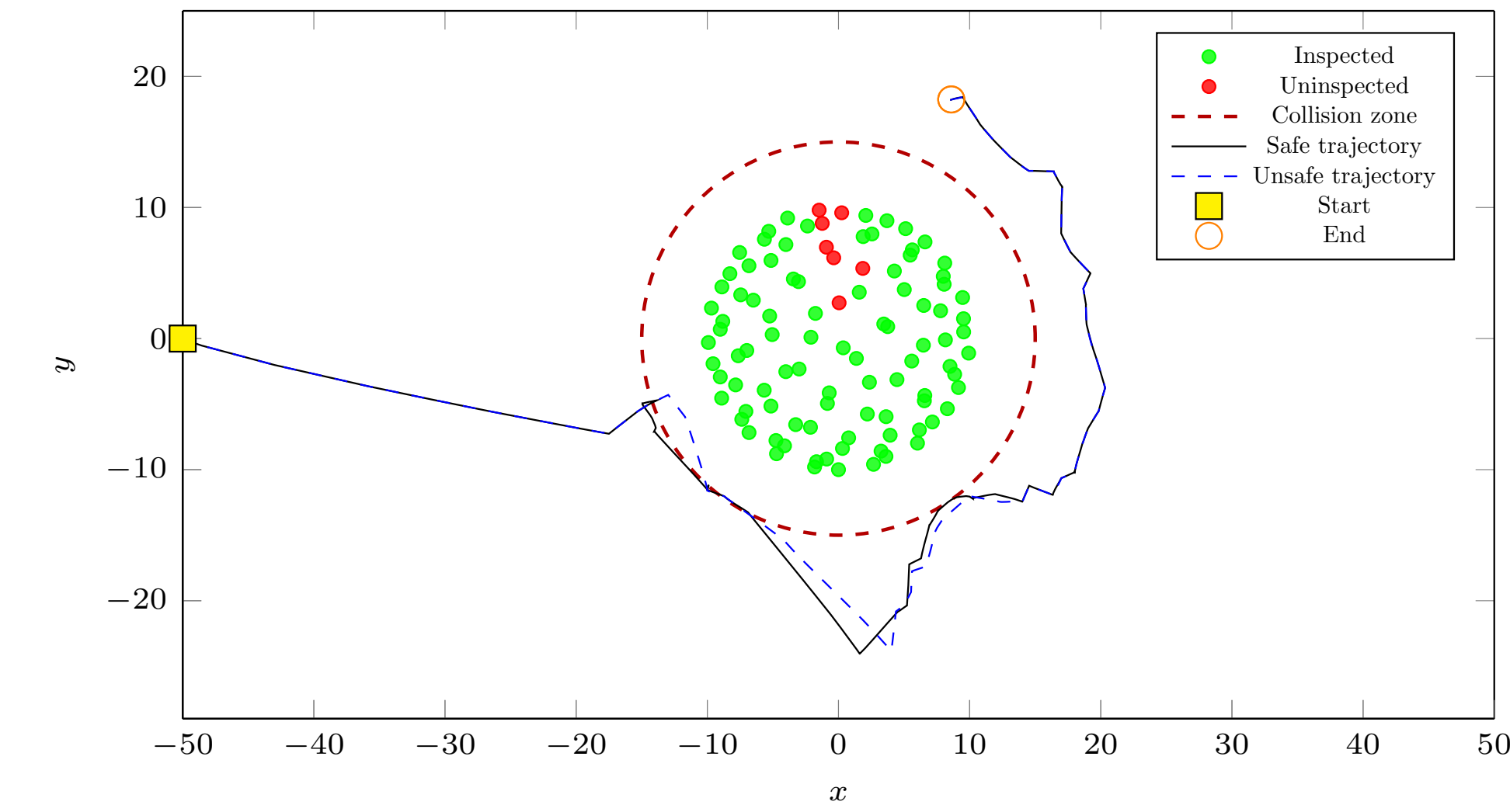


Fig. 2: Trajectory of deputy spacecraft after inspecting the points on the chief satellite for 3000 seconds.

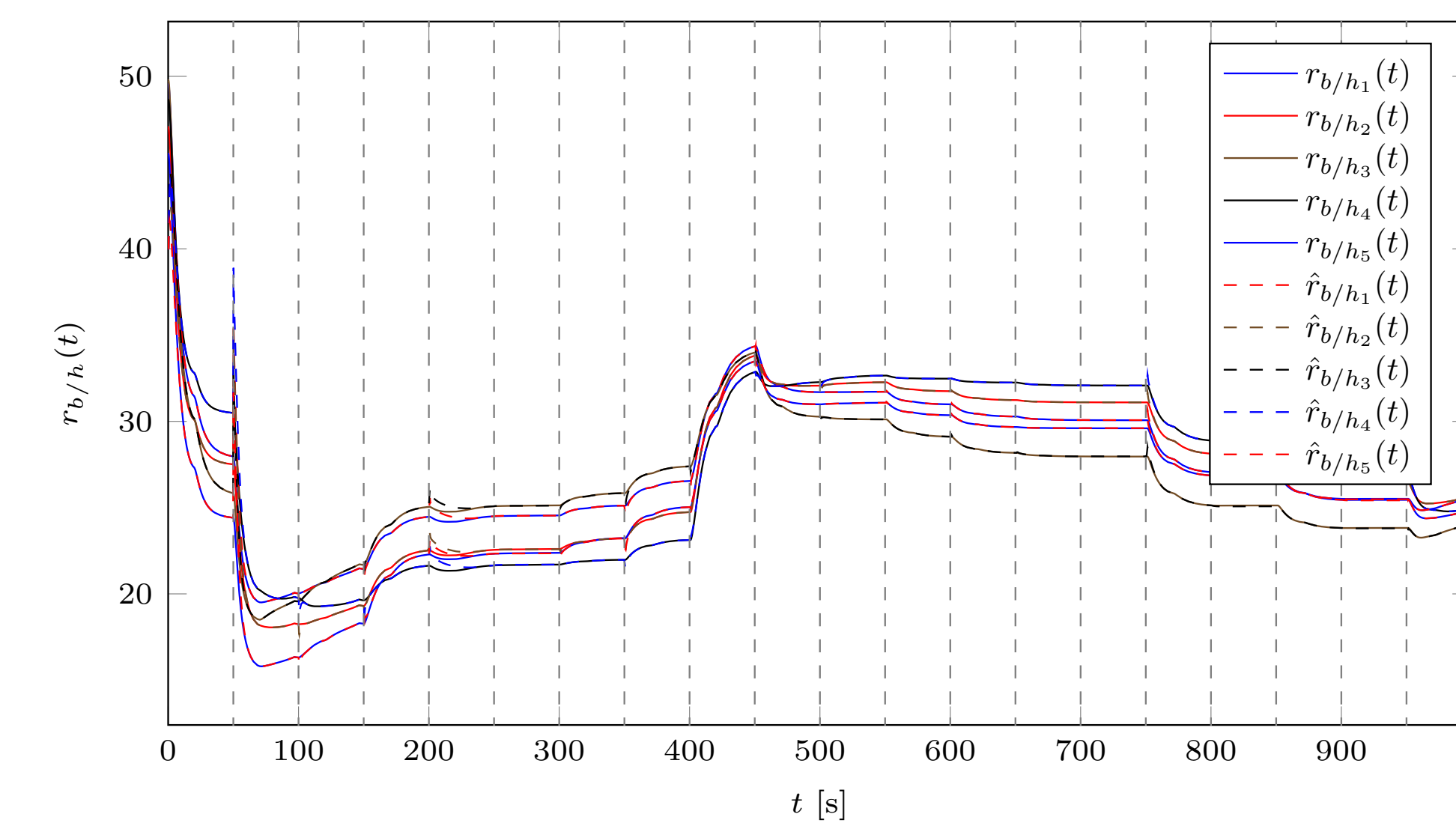


Fig. 3: Trajectories of actual distance and the estimated distance of the deputy spacecraft relative to features on the chief satellite.

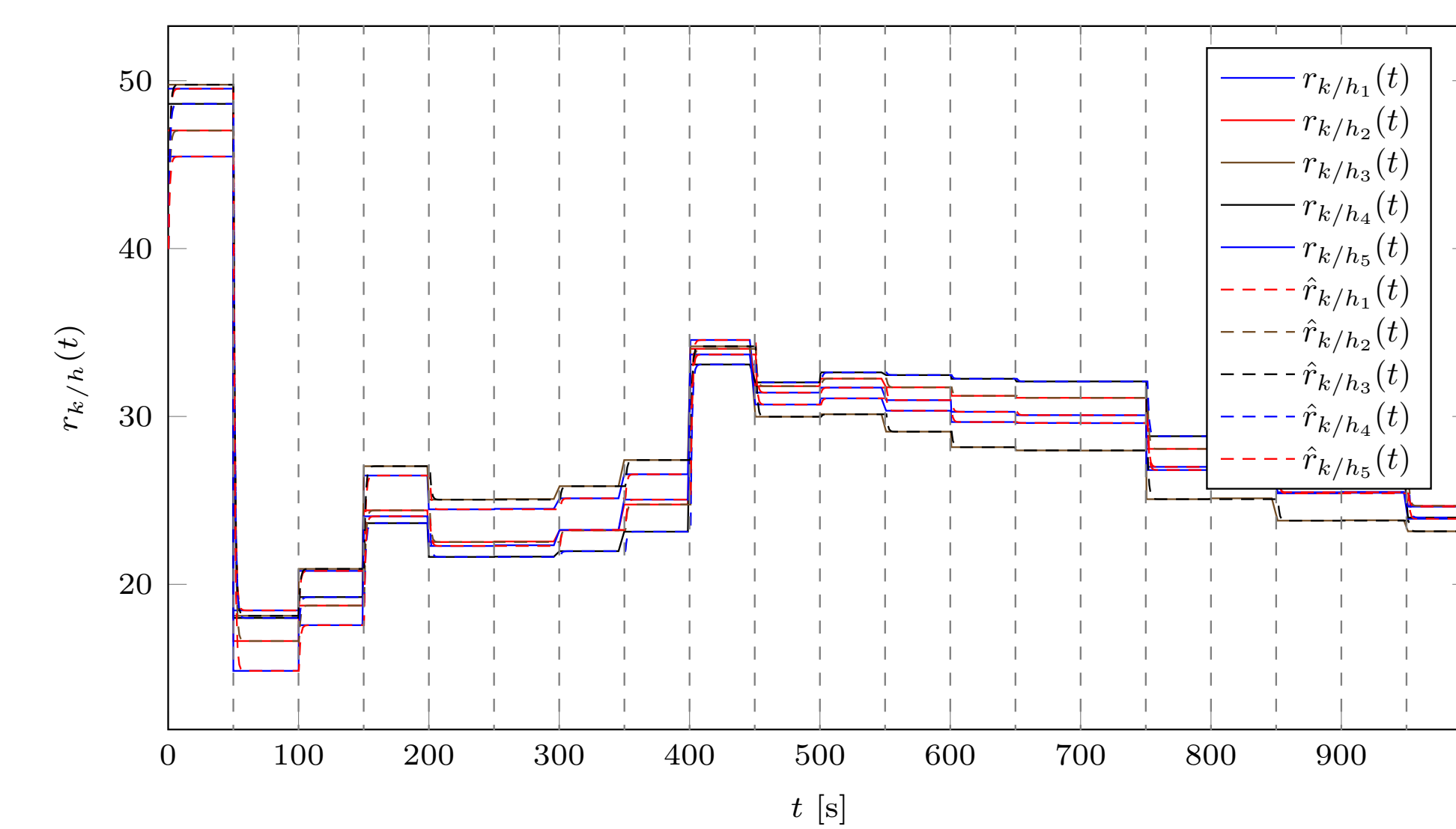


Fig. 4: Trajectories of actual distance and the estimated distance of the key frame relative to features on the chief satellite.

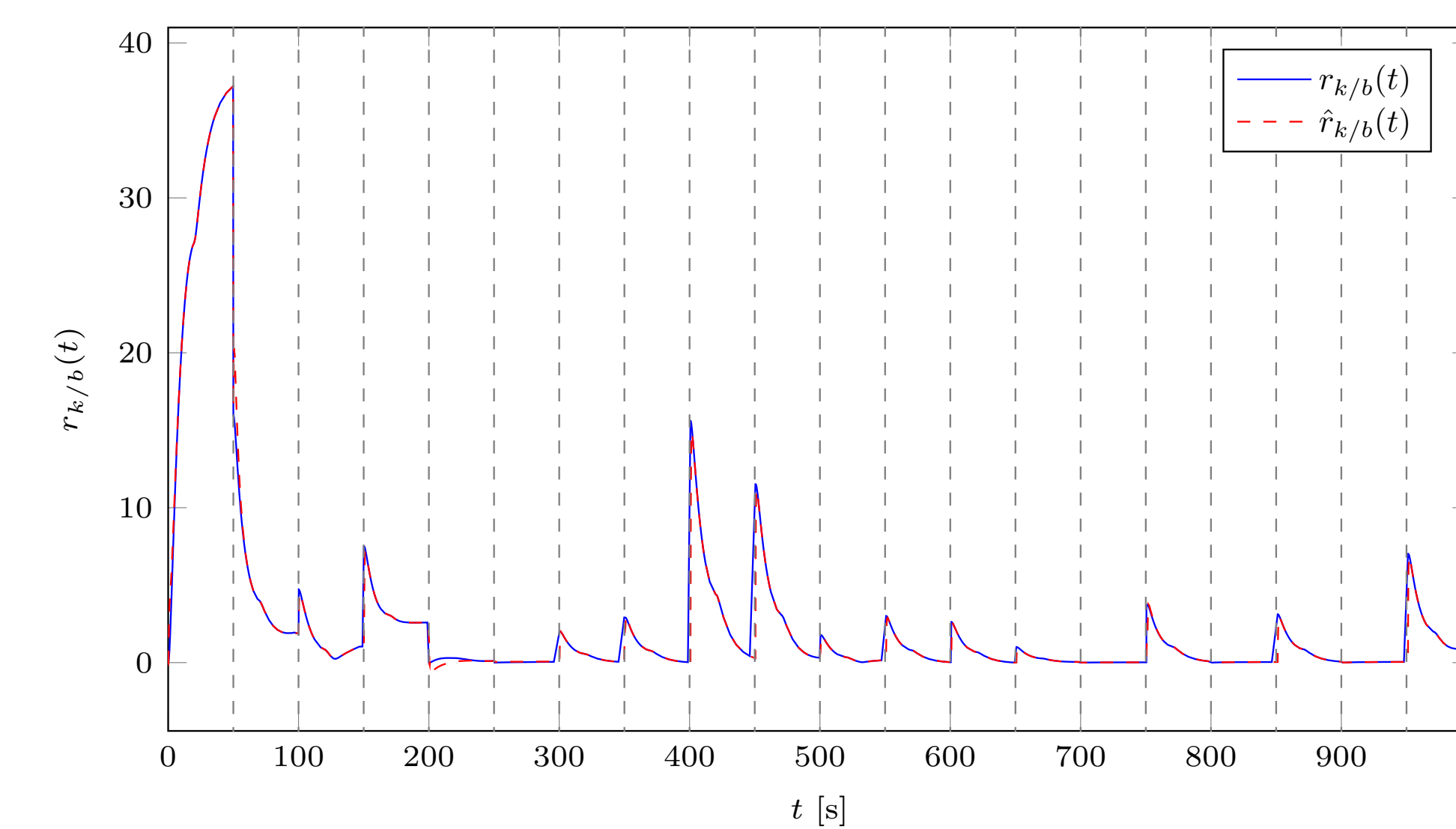


Fig. 5: Trajectories of actual distance and the estimated distance of the keyframe relative to the deputy spacecraft.

Control Design

- The near-optimal control for the deputy spacecraft is designed as

$$\hat{\mathbf{u}}(\hat{\mathbf{x}}(t), t) = -\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P}(t) \hat{\mathbf{x}}(t) - \frac{k_s}{2} \mathbf{R}^{-1} \mathbf{B}^\top \nabla_{\hat{\mathbf{x}}(t)} \Psi^\top(\hat{\mathbf{x}}(t)), \quad (5)$$

where $\hat{\mathbf{x}}(t) \in \mathbb{R}^6$ is the state estimate and $k_s > 0$ is the barrier gain. The Riccati differential equation is $\dot{\mathbf{P}}(t) = -\mathbf{A}^\top \mathbf{P}(t) - \mathbf{P}(t) \mathbf{A} + \mathbf{P}(t) \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P}(t) + \mathbf{Q}$, with terminal condition $\mathbf{P}(t_f) = \mathbf{Q}_f$, where $\underline{\mathbf{Q}}_f = \mathbf{Q}_f + \gamma_c \mathbf{C}^\top \underline{u}_{g/h} \underline{u}_{g/h}^\top \mathbf{C}$ and $\underline{\mathbf{Q}}_f = \underline{\mathbf{Q}}_f^\top \succ 0$.

Analysis

- The image-based observer ensures that the trajectories of the concatenated observer error $\hat{\theta}_i(\cdot)$ is globally uniformly ultimately bounded and the designed control policy ensures $\mathbf{x}(t)$, is locally uniformly ultimately bounded.

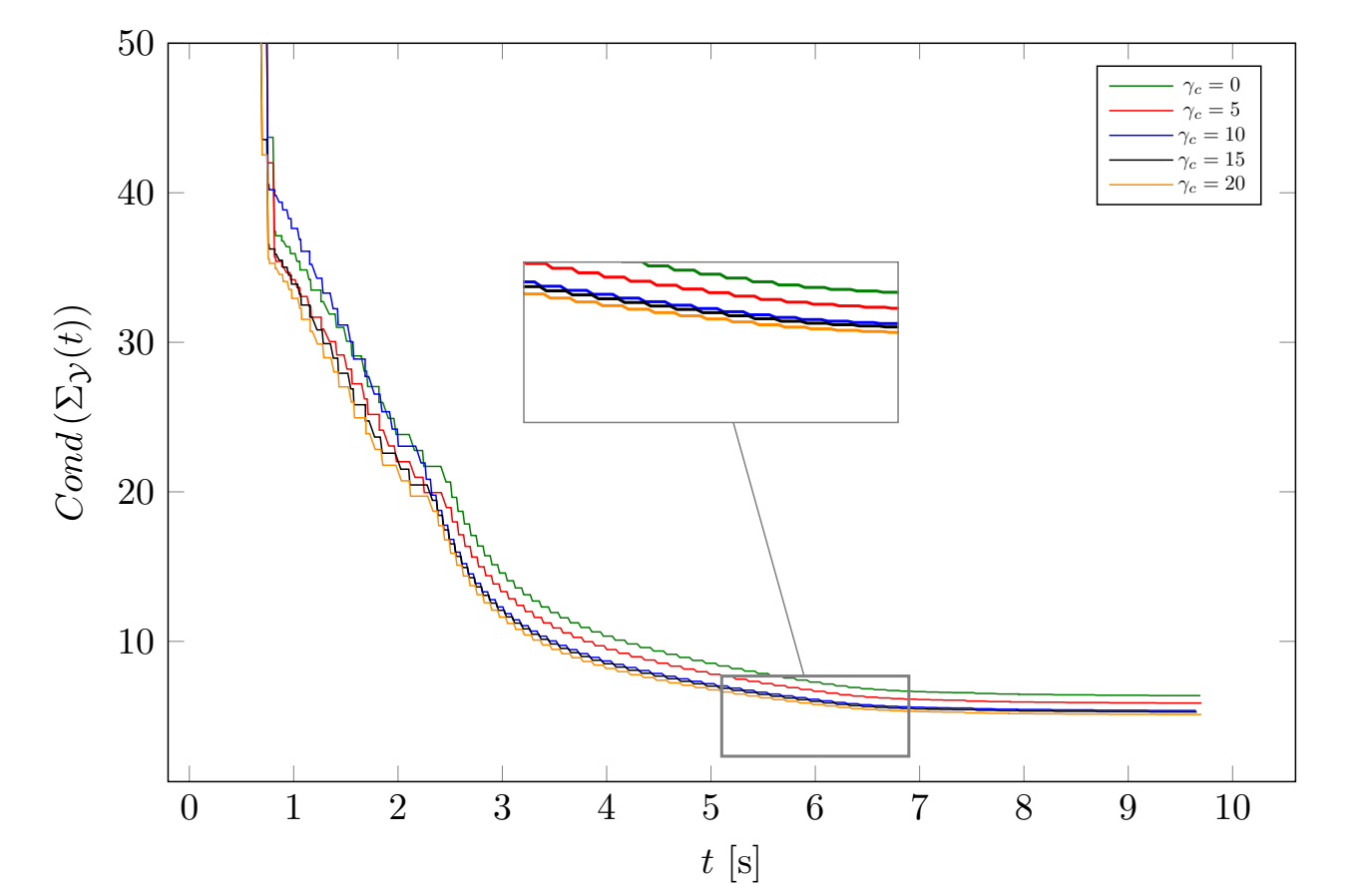


Fig. 6: Condition number of the regressor $\Sigma_{\mathcal{Y}_i}$ for varying orthogonality penalty gains γ_c gain.

Conclusion and Future Work

- Solved on-orbit satellite inspection problem with illumination and FOV constraints.
- Used an image-based distance observer to estimate distances to illuminated features within the deputy spacecraft's FOV.
- Developed a control technique to maximize information gain by improving regressor conditioning.
- Current framework loses information when switching tracked features; future work will explore regressor mixing as a potential solution.

References

- [1] B. Yost and S. Weston, "State-of-the-art small spacecraft technology," Small Spacecraft Systems Virtual Institute, NASA, Tech. Rep., 2024.
- [2] Y. K. Nakka, W. Hönig, C. Choi, A. Harvard, A. Rahmani, and S.-J. Chung, "Information-based guidance and control architecture for multi-spacecraft on-orbit inspection," *J. Guid. Control Dynam.*, vol. 45, no. 7, pp. 1184–1201, 2022.
- [3] B. E. Tweedle and A. Saenz-Otero, "Relative computer vision-based navigation for small inspection spacecraft," *J. Guid. Control Dynam.*, vol. 38, no. 5, pp. 969–978, 2015.
- [4] D. van Wijk, K. Dunlap, M. Majji, and K. L. Hobbs, *Deep reinforcement learning for autonomous spacecraft inspection using illumination*, arXiv:2308.02743, 2023.
- [5] Z. I. Bell, H.-Y. Chen, A. Parikh, and W. E. Dixon, "Single scene and path reconstruction with a monocular camera using integral concurrent learning," in *Proc. IEEE Conf. Decis. Control*, IEEE, 2017, pp. 3670–3675.
- [6] X. Zhao, M. R. Emami, and S. Zhang, "Robust image-based control for spacecraft uncooperative rendezvous and synchronization using a zooming camera," *Acta Astronautica*, vol. 184, pp. 128–141, 2021.

